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A Heterogeneous Data Fusion DNN Framework for SAR-Based Flood Classification: A Case Study of the Muvattupuzha River

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Abstract

Prior knowledge of water resources is essential for accurate flood mapping and disaster assessment. In this study, a hybrid, lightweight, heterogeneous data-fusion framework (HLHDF) is proposed for flood classification by integrating Synthetic Aperture Radar (SAR) imagery with water prior information derived from the WorldCover map and the Global Surface Water Probability (GSWP) bands. The inclusion of these water priors effectively distinguishes between permanent and temporary water bodies, thereby minimizing false detections during flood events. The HLHDF model efficiently learns complex spatial relationships across heterogeneous data sources, thereby improving classification accuracy. The experimental results on the IEEE GRSS DFC 2024 dataset demonstrate that the proposed method achieves an overall F1 score of 87.81%, confirming the advantage of incorporating prior water information with SAR data. As a real-world case study, the proposed framework was evaluated in the Muvattupuzha River basin in Ernakulam district, Kerala, and achieved a flood classification F1 score of 96%, highlighting its robustness and suitability for operational flood mapping. This study highlights the potential of deep learning–based data fusion frameworks to enhance flood mapping reliability and support timely disaster response and management efforts.

Impact Statement

Most flood change-detection studies rely exclusively on SAR imagery, without incorporating the geographical characteristics of the study area. As a result, region-specific information, such as land-cover patterns and historical hydrological behavior, is often overlooked. Furthermore, the probability of water occurrence in a particular region is rarely considered in the literature. The fusion of multisource geospatial data enhances situational awareness during extreme weather events, enabling faster and more informed decision-making. To address these limitations, this study presents a hybrid lightweight heterogeneous data-fusion (HLHDF) framework for flood mapping that integrates SAR imagery with prior water information to significantly improve disaster monitoring accuracy. By effectively distinguishing between permanent and temporary water bodies, the proposed approach reduces false flood detection, thereby overcoming a major limitation of conventional flood mapping methods. The high F1-score (87.81%) on the IEEE GRSS DFC 2024 dataset and real-world validation over the Muvattupuzha River Basin (96% F1-score) demonstrate the model's robustness, reliability, and operational readiness for practical flood monitoring applications. This framework has strong potential to support disaster management authorities in early warning systems, emergency planning, and resource allocation. Moreover, the

methodology can be adapted to other flood-prone regions globally, contributing to resilient infrastructure planning and climate change adaptation strategies. Overall, the proposed model advances intelligent flood risk assessment, promoting safer communities and more effective disaster response mechanisms.

Contents

1 Introduction **2**
2 Related Work **2**
3 Methodology **3**
3.1 Preprocessing 3
3.2 Feature Vector Generation 4
3.3 Deep Neural Network Model 4
4 Result and Discussion **4**
4.1 Performance Evaluation on Benchmark Dataset 4
4.2 Real-World Case Study: Flood Mapping over the Muvattupuzha River Basin 5
5 Conclusion **8**

1. Introduction

Floods have become among the most devastating and widespread natural disasters of the current decade, primarily because of their sudden onset and severe impact on human life, infrastructure, and socio-economic activities [26]. The rapid onset of flooding often leads to extensive loss of life and damage to essential resources [32]. Accurate and timely water mapping plays a vital role during flood events, serving as a key process in disaster management and mitigation [11, 23]. Effective floodwater delineation not only supports emergency response and resource allocation but also significantly safeguards human lives and minimizes the overall impact of such disasters [1, 31].

Over the years, machine learning methods have evolved from traditional statistical and rule-based approaches to highly data-driven models capable of capturing complex spatial–temporal relationships [36]. Initially, algorithms such as decision trees and random forests were widely used for flood detection due to their simplicity, interpretability, and ability to handle heterogeneous geospatial data [28]. However, their limited generalization capability under dynamic environmental conditions motivated the adoption of more sophisticated learning models [22].

The evolution from neural networks to deep neural networks (DNNs) has enabled automatic feature learning and improved representation capabilities [10]. Recently, pretrained models such as DeepLabV3+ and SegFormer have shown strong performance in semantic segmentation and flood mapping [8, 16]. When combined with multi-source data fusion—including SAR backscatter, DEMs, and water probability maps—these models offer a robust framework for high-precision flood classification and disaster management [35]. However, challenges remain due to SAR-specific issues such as speckle noise, complex scattering, and dynamic scene variations [13].

To overcome these limitations, the proposed methodology integrates the Global Surface Water probability(GSWP) band with the ESA WorldCover map to enhance the accurate identification of water pixels from SAR imagery using DNN. This fusion strategy effectively mitigates misclassification issues and improves the reliability of flood extent delineation, thereby contributing to more accurate and impactful flood mapping and management outcomes.

2. Related Work

Flood classification methods can be grouped into three main categories: hyperspectral, SAR-based [4], and UAV/drone-based approaches [15]. Hyperspectral methods [14] use water-sensitive indices and

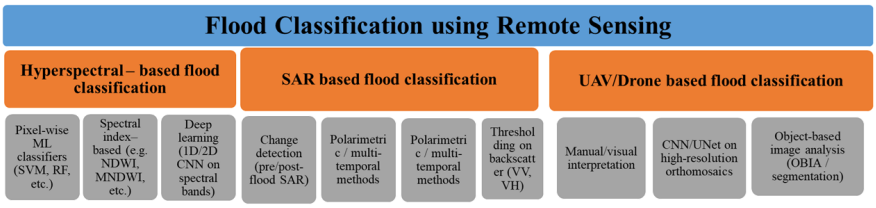


Figure 1: Conceptual framework summarizing related works in remote sensing–based flood classification.

machine learning classifiers, while SAR-based techniques—most widely used—employ thresholding, change detection, polarimetric features, and deep learning. UAV and drone platforms [2] provide high-resolution optical imagery for detailed flood mapping through object-based analysis and semantic segmentation. These approaches form the basis of remote sensing-based flood mapping (Fig. 1).

The analysis of synthetic aperture radar (SAR) imagery has long been recognized as a challenging task because to the inherent complexity of radar backscatter and the nonlinear interactions between signals and surface structures [20]. Unlike optical or hyperspectral imagery, SAR data are highly sensitive to surface roughness, moisture content, and incidence angles, making direct interpretation or threshold-based classification unreliable [29]. The presence of speckle noise, geometric distortions, and mixed-pixel effects further complicates accurate flood delineation [5].

To address these challenges, various machine learning (ML) and deep learning (DL) methods have been applied to SAR-based flood mapping [4]. Traditional classifiers such as SVM, random forest, and gradient boosting have been used with dual-polarized Sentinel-1 data [3, 12], but they rely on handcrafted features and show limited robustness for large or heterogeneous datasets. In contrast, deep neural networks learn hierarchical features directly from imagery, enabling better discrimination of complex land-cover patterns and improving flood classification performance [7, 25, 30]. These advances form the basis of modern DL-based remote sensing analysis.

In addition to single-source analysis, data fusion has gained attention to overcome the limitations of SAR-only classification. Integrating SARs with optical imagery, DEMs, global surface water probabilities, and ESA WorldCover enhances feature diversity and semantic understanding [18, 21]. This fusion improves discrimination between flooded, non-flooded, and permanently inundated areas [6].

In summary, while deep learning and data fusion approaches have substantially improved SAR-based flood detection, challenges remain in achieving consistent generalization across diverse landscapes. This study extends the methodologies presented in [34] and [35] by developing a hybrid, lightweight, heterogeneous data-fusion framework (HLHDF) that combines SAR polarizations (VV, VH) with global ancillary data, including global surface water probability (GSWP) and ESA WorldCover layers, thereby achieving enhanced spatial precision and classification reliability for flood mapping.

3. Methodology

The overall workflow of the proposed approach involves three major stages: *preprocessing*, *feature vector generation*, and *flood classification* via a deep neural network (DNN) model.

3.1. Preprocessing

Synthetic aperture radar (SAR) images are inherently affected by speckle noise, which is multiplicative noise resulting from the coherent nature of radar backscattering. This noise appears as fine-grained random patterns that degrade the visual quality and complicate classification tasks. To mitigate this effect, the speckle reducing anisotropic diffusion (SRAD) [19] technique is applied. SRAD performs

adaptive diffusion that preserves edges and structural details while effectively suppressing speckle in homogeneous regions.

The observed SAR image intensity $I_{x,y}$ can be modeled as:

$$I_{x,y} = S_{x,y} \cdot n_{x,y} \tag{3.1}$$

where $S_{x,y}$ represents the true backscatter coefficient, and $n_{x,y}$ is the multiplicative speckle noise.

3.2. Feature Vector Generation

After noise suppression, the SAR data were converted into feature vectors. Each pixel was represented by VV and VH backscatter values, along with auxiliary features from the *global surface water probability (GSWP)* layer and the *ESA WorldCover map*. These additional features improve class separability, especially in areas with mixed or uncertain backscatter signatures.

Formally, each pixel p_i is expressed as:

$$x_i = \left[VV_i,\ VH_i,\ ESA_i,\ GSWP_i\right] \tag{3.2}$$

where $\mathbf{x}_i$ is the input feature vector for the DNN classifier.

3.3. Deep Neural Network Model

The proposed DNN model consists of one input layer, two hidden layers with ten neurons each, and an output layer performing binary flood classification. The layers utilize the rectified linear unit (ReLU) activation function:

$$fz = \max\left(0, z\right) \tag{3.3}$$

to introduce non-linearity and accelerate convergence. The output layer applies a sigmoid activation function to estimate the flood probability $\hat{y} \in \left[0, 1\right]$.

Model parameters were optimized through backpropagation via the Adam optimizer (Adaptive Moment Estimation) [9].

The model was trained on a very limited dataset consisting of 10 images, while the remaining 1621 images were used exclusively for testing. A feedforward deep neural network with two hidden layers (10 neurons each, ReLU activation) and a softmax output layer was trained using the Adam optimizer for 100 epochs with a mini-batch size of 62. The visualization of the proposed work is illustrated in Figure 2.

4. Result and Discussion

The performance of the proposed method was first evaluated on a benchmark dataset to establish its effectiveness under controlled conditions. The trained model was subsequently applied to a real-world flood scenario to assess its generalizability and practical applicability. This two-stage evaluation framework enables a comprehensive analysis of the model's robustness and reliability across both standard and real-world settings.

4.1. Performance Evaluation on Benchmark Dataset

The IEEE GRSS DFC 2024 dataset [24] is used to evaluate the proposed heterogeneous data fusion approach. It comprises multi-sensor data, including Sentinel-1 SAR (10 m resolution), ESA WorldCover (10 m resolution), and global surface water probability (GSWP, 10 m resolution), with image patches

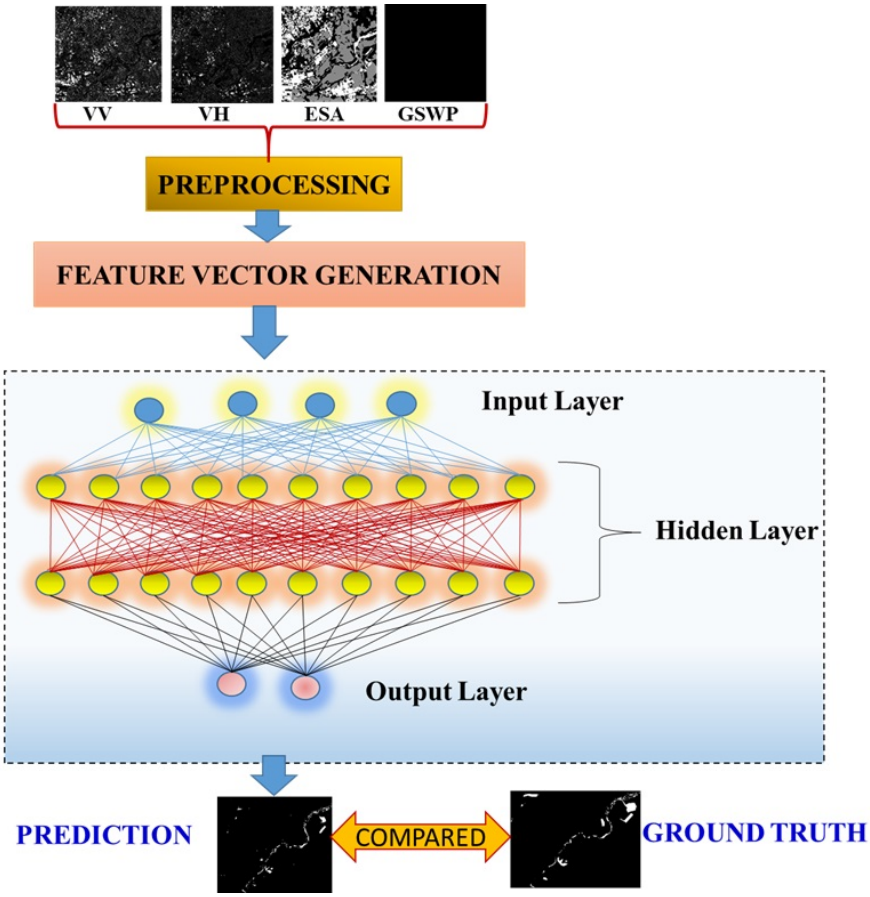


Figure 2: Visualization of the proposed methodology illustrating the sequential workflow — from data preprocessing to model training, classification, and final output generation.

of size 512 × 512 pixels. In this study, VV and VH polarization bands were utilized to capture time-dependent backscatter variations. The ESA WorldCover map provides static land cover information (2020), whereas the GSWP layer represents the long-term probability of water occurrence.

Table 1 compares flood classification performance with and without auxiliary data. Using only VV and VH achieved 81.62% accuracy, whereas adding ESA WorldCover and global surface water probability (GSWP) individually improved the performance to 85.79% and 84.50%, respectively [35]. The combined integration (VV+VH+GSWP+ESA) produced the best results, with 86.28% accuracy and an 87.81% F1-score, demonstrating the robustness of the proposed framework. Further comparison with state-of-the-art methods (Table 2) and visual results (Fig. 3) show that the proposed model generates more spatially coherent flood maps with improved boundary delineation and fewer misclassifications. By fusing SAR data with global surface water probability and land-cover data, the model better distinguishes permanent and temporary water bodies, significantly outperforming single-source approaches.

4.2. Real-World Case Study: Flood Mapping over the Muvattupuzha River Basin

The study area is located in the eastern Ernakulam district, Kerala, around Muvattupuzha, and is bounded by four corner coordinates (see Figure 4). It spans the midland and highland zones of central

Table 1: Comparison of flood classification performance metric obtained with and without the integration of the water probability band and ESA WorldCover map.

Bands	**Accuracy%**	**Precision%**	**Recall%**	**F1score%**
VV+VH	81.62	96.49	74.38	84
VV+VH+GSWP	84.50	97.30	77.47	86.26
VV+VH+ESA[35]	85.79	98.38	78.59	87.38
VV+VH+GSWP+ESA	**86.28**	**98.87**	**78.98**	**87.81**

Table 2: Performance Comparison of the Proposed Model with Existing Models on the Same Dataset

Model Used	Bands	F1-score
Naive CNN[27]	All 6 bands	80.9%
Random Forest[33]	VV+VH	82%
U-NET[27]	VV+VH+CDEM+WaterProb+WBM +DEM-WB+Distance,VV/VH	82.4%
Multi-UAFNet+Post-Proc.[17]	VV+VH	82.99%
Segformer[27]	VV+VH+MDEM+CDEM+Water Probail-ity+WBM+Distance+VV/VH	83.8%
DNN	VV+VH	84%
Random Forest [34]	VV+VH+GWSP	85.71%
DNN[35]	VV+VH+ESA	87.38%
DNN(Our)	**VV+VH+GSWP+ESA**	**87.81%**

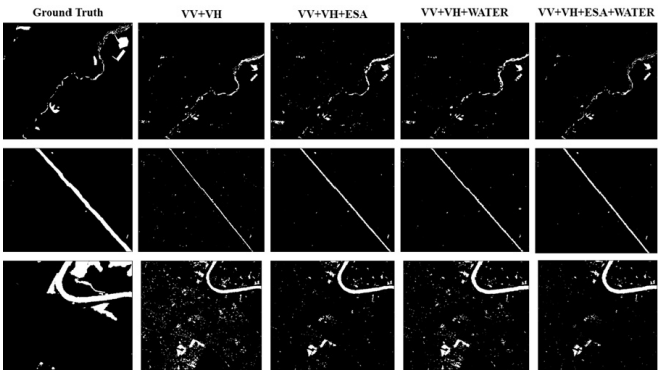


Figure 3: Visual Comparison of Our Model's Results with Other Heterogeneous Data Fusion Approaches.

Kerala and is mainly drained by the Muvattupuzha River and its tributaries, making it highly prone to monsoon flooding. The region receives heavy southwest monsoon rainfall, exceeding 2500 mm annually. Following extreme rainfall and severe flooding in August 2018, Sentinel-1 SAR imagery from 22 August 2018 was used to capture post-flood conditions. The area's diverse land cover and complex drainage network make it suitable for evaluating flood mapping techniques.

The multi-source geospatial data were processed via the Google Earth Engine (GEE). The area of interest was defined via a user-drawn polygon (512x512), and Sentinel-1 SAR data (21–22 August

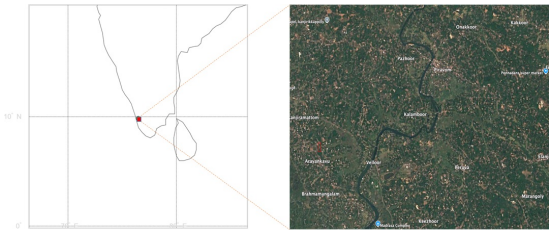
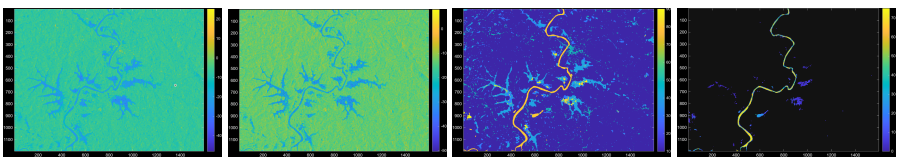


Figure 4: Spatial location of the study area with latitude–longitude grid and major reference landmarks.

2018) were selected from the COPERNICUS/S1_GRD collection in IW (Interferometric Wide Swath) mode with descending orbits and dual polarization (VV, VH). Standard preprocessing methods such as thermal noise, radiometric calibration, terrain correction, and resampling to 10 m spatial resolution were used. After that, median compositing was used to eliminate speckle noise. Auxiliary datasets (ESA WorldCover v200 and JRC Global Surface Water) were resampled, clipped, and stacked with VV and VH to form a four-band input image (Fig. 5).



(a) VV (b) VH (c) World cover map (d) GSWP

Figure 5: Multi-source datasets used in the analysis: (a) Sentinel-1 VV backscatter, (b) Sentinel-1 VH backscatter, (c) ESA WorldCover v200 land-cover classification, and (d) JRC Global Surface Water occurrence (1984–2021), representing long-term water history.

Table 3 summarizes the statistics of the four feature bands (VV, VH, WorldCover, and GSWP). GSWP missing values (NaN), indicating areas without historical water records, were set to 0, and a binary validity mask was added as a fifth band. All bands were normalized to [0,1], forming a standardized five-band input. Four model configurations (VV+VH, VV+VH+ESA, VV+VH+GSWP, and VV+VH+ESA+GSWP) were evaluated to quantify the contribution of ancillary data. A reference water mask, derived by combining permanent GSWP and Otsu-based SAR water pixels, was used for performance assessment.

	VV	VH	ESA	GSWP
Min	-48.0835	-50.2998	10	0
Max	25.2571	7.9522	90	75
NaN Count	0	0	0	244271

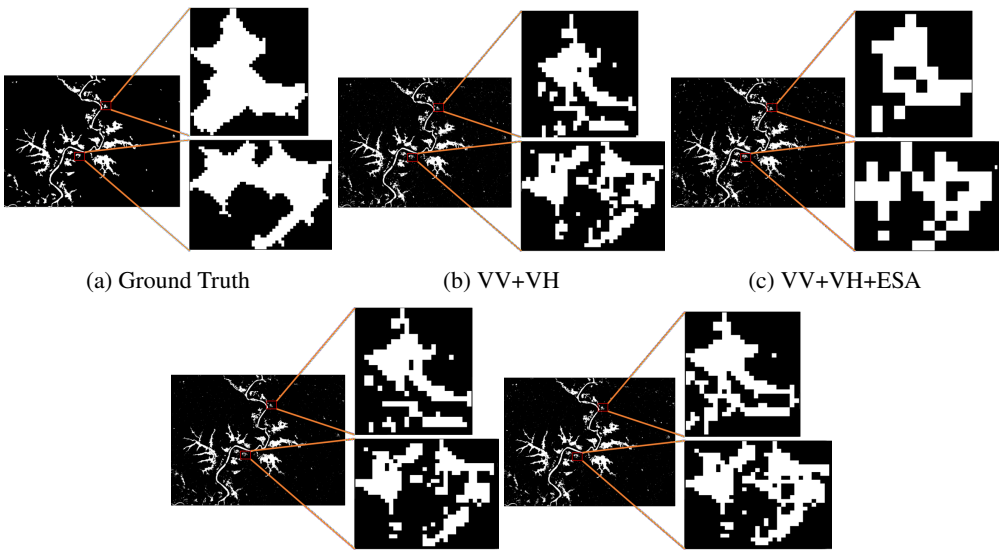
Table 3: Descriptive statistics of the four feature bands used in the study, including the observed minimum and maximum pixel intensities and the corresponding NaN occurrence for each band.

An ablation study (Table 4) assessed the impact of ESA and GSWP data. The SAR-only baseline (VV+VH) achieved strong performance (95.07% accuracy, 95.27% F1-score). Adding ESA (VV+VH+ESA) improved results (95.65% accuracy, 95.81% F1-score), highlighting the value of land-cover information. Using GSWP alone slightly reduced recall (89.16%), indicating limited standalone

benefit. However, combining ESA and GSWP (VV+VH+ESA+GSWP) produced the best performance (96.16% accuracy, 96.28% F1-score), confirming their complementary contribution. Figure 6 shows the study area and zoomed outputs from the four model configurations. This highlights how integrating VV, VH, ESA WorldCover, and GSWP improves land–water boundary delineation, supporting accurate flood classification.

Band Used	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
VV+VH	95.07	99.47	91.42	95.27
VV+VH+ESA	95.65	99.49	92.4	95.81
VV+VH+GSWP	93.76	99.63	89.16	94.11
VV+VH+ESA+GSWP	**96.16**	99.36	93.39	**96.28**

Table 4: Performance comparison of the four models, illustrating the results generated for the study area



(a) Ground Truth (b) VV+VH (c) VV+VH+ESA

(d) VV+VH+GSWP (e) VV+VH+ESA+GSWP

Figure 6: Zoomed-in visualization of the reference ground truth and the four classification outputs derived from distinct feature combinations (VV+VH, VV+VH+ESA, VV+VH+GSWP, and VV+VH+ESA+GSWP), illustrating the model performance across the selected region.

5. Conclusion

This study demonstrates the effectiveness of integrating water probability and ESA WorldCover data with SAR imagery for improved flood classification. The proposed hybrid lightweight heterogeneous data-fusion model outperforms existing approaches, achieving an F1-score of 87.81%. The inclusion of auxiliary datasets enhanced spatial consistency and boundary delineation, reducing the misclassification of non-flooded areas. Comparative results confirm that heterogeneous data fusion significantly improves flood detection accuracy. An evaluation of a region with complex flood dynamics further highlights

the robustness and generalizability of the proposed method. Future work will extend this framework to multi-temporal data and additional geographic regions to assess its scalability and adaptability.

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Competing Interests. The authors declare that they have no competing interests relevant to the content of this article.

Data Availability Statement. The code and regional dataset supporting this study are publicly available at https://doi.org/10.5281/zenodo.19200333. Furthermore, benchmark data from the IEEE GRSS Data Fusion Contest 2024 dataset are also publicly available.

Ethical Standards. The research meets all ethical guidelines, including adherence to the legal requirements of the study country.

Author Contributions. Conceptualization: Jayasree Thazhath Veedu; Rajesh Reghunadhan. Methodology: Jayasree Thazhath Veedu; Rajesh Reghunadhan. Data curation: Jayasree Thazhath Veedu. Data visualisation: Jayasree Thazhath Veedu. Writing original draft: Jayasree Thazhath Veedu; Rajesh Reghunadhan. All authors approved the final submitted draft.

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